

time horizons. This is the heart of the sustainable spending problem: spend more, and your money will not last as long. But these fundamentally important links are easy to lose sight of later when we enter the world of the “4 percent rule” for retirement spending.

With these functions, the task becomes to determine appropriate values for the underlying inputs. How much do we want to be able to spend? How long should our spending strategy work? What is a reasonable rate of return assumption for our portfolio? How much should we try to leave at the end of the time horizon? We have to answer these questions within the context of retirement risk: how likely is it that we outlive our planning age, and what happens if the market returns are less than we assumed?

The answer, invariably, is that spending will have to be reduced at some point later in retirement. We must also consider the impact of a potential spending reduction in terms of how much it lowers our standard of living. With enough income sources from outside the portfolio and with greater flexibility to adjust spending in response to market performance, we may be more willing to accept potential late-in-life reductions for our portfolio spending.

Returning to an earlier example, we saw that a 7 percent rate of return would allow for a 7.53 percent initial spending rate for a thirty-year retirement. That was the state of analysis in the mid-1990s when William Bengen came onto the scene and his research led him to coin the 4 percent spending rule. If the long-term average real compounding return from the stock market is 7 percent, does that really mean you can safely use a 7.5 percent withdrawal rate from a 100 percent stocks portfolio without worrying about running out of wealth for thirty years? No, it does not.

Stocks do not earn their average real return each year. Some years they go up, and some years they go down, as recent investors know all too well. For a retiree who is taking distributions from savings, the sequence of market returns matters. If a retiree’s portfolio drops in value during the early retirement period, portfolio withdrawals will dig a further hole. Climbing out of this hole becomes increasingly difficult, even if a subsequent market recovery arrives. This is sequence-of-returns risk. Sequence risk amplifies